

From Prestige to Presence: Algorithmic Visibility and Citation Bias in the Age of Generative AI

Dag Øivind Madsen, University of South-Eastern Norway

Shahab Saquib Sohail, VIT Bhopal University

Abstract

This paper examines how generative artificial intelligence (GenAI), particularly large language models (LLMs), is reshaping scholarly citation patterns through mechanisms of algorithmic visibility, open access bias, and systemic risk. Building on established research on the open access (OA) citation advantage, we argue that GenAI systems amplify this effect by structurally privileging sources that are freely available and well-indexed on the open web, while sidelining closed-access or under-indexed scholarship. We introduce a citation visibility funnel model to show how accessibility, absorption, and credibility filters shape which publications become part of the “citable canon.” We then analyze how megajournals and predatory outlets may exploit this dynamic, raising the risk of citation contamination. To guide future inquiry, we propose seven propositions organized into three clusters—algorithmic visibility, researcher behavior, and systemic risks & inequalities—summarized in a research roadmap. Together, these contributions highlight how GenAI may both democratize and distort scholarly communication, calling for stronger transparency in AI training data, safeguards against low-rigor publishing, and critical awareness of algorithmic mediation in academic evaluation.

Keywords: Generative AI; Scholarly Citation; Algorithmic Visibility; Open Access; Citation Bias; Megajournals; Predatory Publishing; Epistemic Risk; Academic Publishing; Matthew Effect

1 Introduction

In recent decades, the scholarly communication system has undergone profound changes. The transition from print to digital publishing (Roberts, 1999), the growth of open access (OA) platforms (Laakso et al., 2011), and the increasing reliance on citation metrics in research evaluation (Moed, 2005) have all contributed to a complex, data-rich ecosystem in which visibility, legitimacy, and impact are closely intertwined. Now, a new transformation is underway: the integration of generative artificial intelligence (GenAI) into academic workflows (Carobene et al., 2024; Khalifa & Albadawy, 2024). Large language models (LLMs) such as ChatGPT, Claude, and Gemini are being adopted across disciplines to assist with writing, summarization, ideation, and increasingly, literature discovery (Khalifa & Albadawy, 2024). As these tools become normalized, a pressing question emerges: How might the design of these systems—and their training on skewed datasets—reshape what is cited, elevated, or overlooked in academic research? As citation becomes increasingly automated, the criteria for inclusion in scholarly dialogue may shift from rigor to reach, from prestige to presence. In this paper, we explore whether the patterns emerging in GenAI-mediated citation reflect a distinct evolution of Merton’s original formulation—a Matthew Effect 2.0, grounded not in academic prestige but in algorithmic visibility.

This paper addresses that question by examining the intersection between generative AI and the well-documented OA citation advantage. During the 2000s, numerous studies showed that articles freely accessible online tend to receive more citations than those locked behind paywalls, a trend attributed to greater visibility and ease of access (Antelman, 2004; Eysenbach, 2006a, 2006b; Lawrence, 2001; Norris et al., 2008). This OA citation advantage has continued to pique the interest of researchers (Huang et al., 2024; Langham-Putrow et al., 2021; Piwowar et al., 2018; Wang et al., 2015), especially as the growth of OA publishing has been very strong (Hook et al., 2019; Kramer, 2017).

We argue that generative AI may significantly amplify this OA advantage, not simply by facilitating access, but by structurally favoring OA content in the knowledge outputs it provides. Because LLMs are trained on corpora drawn largely from the open internet and publicly available texts, their internal knowledge—and their reference suggestions—are inherently biased toward sources that are freely accessible. Closed-access journals, even those with high prestige and rigorous peer review, are underrepresented in these models unless specific licensing agreements are in place.

As a result, researchers using AI to assist with literature review or citation generation may be more likely to encounter and cite openly available articles, including those from lesser-known, lower-ranked, or even questionable outlets. This dynamic raises a series of critical issues for academic publishing and evaluation. First, it could reconfigure the prestige economy of scholarship by shifting citation flows away from elite journals and toward OA megajournals or preprint repositories. Second, it introduces epistemic risks, as visibility rather than rigor may

increasingly determine what knowledge is cited and disseminated. Third, it may exacerbate the influence of predatory publishing models, which thrive on openness but often lack proper editorial standards (Eriksson & Helgesson, 2017; Mills & Inouye, 2021).

Through a conceptual and interdisciplinary lens, this paper examines the implications of generative AI for scholarly visibility and citation patterns. We begin by reviewing the established literature on the OA citation advantage and exploring how GenAI systems are constructed and used in academic contexts. We then outline the mechanisms through which AI may amplify OA citation patterns, paying particular attention to the role of algorithmic visibility and epistemic filtering. In the second half of the paper, we address the risks posed by predatory publishing and the increasing role of OA megajournals in AI-mediated citation ecosystems. We conclude by proposing a research agenda to guide future empirical work and policy development in this emerging area.

2 Algorithmic Visibility and the Matthew Effect

2.1 The Open Access Citation Advantage

The notion that OA publications receive more citations than paywalled ones is a well-established finding in bibliometric research. Early studies attributed this phenomenon to greater discoverability and ease of access: articles that can be read without institutional credentials are more likely to be found, read, and cited. Piwowar et al. (2018), in a large-scale analysis concluded that OA articles receive 18% more citations on average than their closed-access counterparts. More recently, Huang et al. (2024) extended this work, demonstrating that OA outputs not only attract more citations, but are also cited by a more diverse set of authors across disciplines, geographies, and institutional contexts.

While this citation advantage spans fields, its magnitude varies—and may be further amplified or distorted by algorithmic filters. In some cases, the difference is marginal, while in others—particularly in fields with strong preprint cultures—the effect is pronounced. Importantly, the OA citation advantage is not solely a function of access; it is also shaped by the structure of discovery tools such as Google Scholar, indexing databases, and increasingly, algorithmic intermediaries.

2.2 Algorithmic Reinforcement of Visibility

Generative AI introduces a new layer to this dynamic. LLMs such as ChatGPT and Claude are trained on corpora that include large volumes of freely available text from the internet. While the specific training datasets are proprietary, it is widely acknowledged that these models draw heavily from sources like Wikipedia, arXiv, PubMed Central, and open-access journal content. Paywalled journal articles, in contrast, are typically excluded unless licensing agreements allow access to full-text content. As a result, the model's internal knowledge is weighted heavily

toward OA sources. When users prompt an LLM to summarize a research area or suggest relevant references, the outputs are disproportionately influenced by what the model has seen—namely, what was accessible during training.

In effect, the OA citation advantage becomes algorithmically reinforced. AI-generated literature summaries, research assistance, and citation suggestions are more likely to favor OA articles, not because they are necessarily better or more rigorous, but because they are more visible to the AI. This represents a shift from a user-driven discovery model (e.g., database searching) to an AI-mediated selection model, where the available universe of sources is pre-filtered by training data or retrieval mechanisms that favor openness.

The implications of this shift are considerable. First, it may accelerate the already growing influence of open-access journals, including new and relatively untested outlets. Second, it may reduce the relative citation share of established but closed-access journals, even in cases where they publish higher-quality or more impactful research. Third, it may lead to a feedback loop in which articles that are more visible to AI become more cited, and thus more visible again—amplifying a kind of algorithmic Matthew Effect grounded not in prestige, but in accessibility. This resonates with altmetrics research, which finds that digital attention and discoverability are shaped by visibility-driven interactions rather than rigorous evaluation (Haustein et al., 2016). This “Matthew Effect 2.0” reflects how cumulative advantage operates in an AI-mediated context: articles that are freely available and well-indexed are more likely to be included in LLM training data, which makes them more likely to be surfaced in response to future prompts. Over time, this creates a self-reinforcing cycle where visibility begets visibility—not necessarily because of scholarly rigor, but because of algorithmic access. Over time, this dynamic may not only alter what is visible but also reshape perceptions of prestige itself.

2.3 Citation Visibility Funnel

The mechanisms discussed above can be represented in a visual model. Figure 1 illustrates how citation visibility narrows through successive stages of accessibility, absorption, and credibility. At each stage, structural biases may be amplified, filtering which sources ultimately enter the “citable canon.”

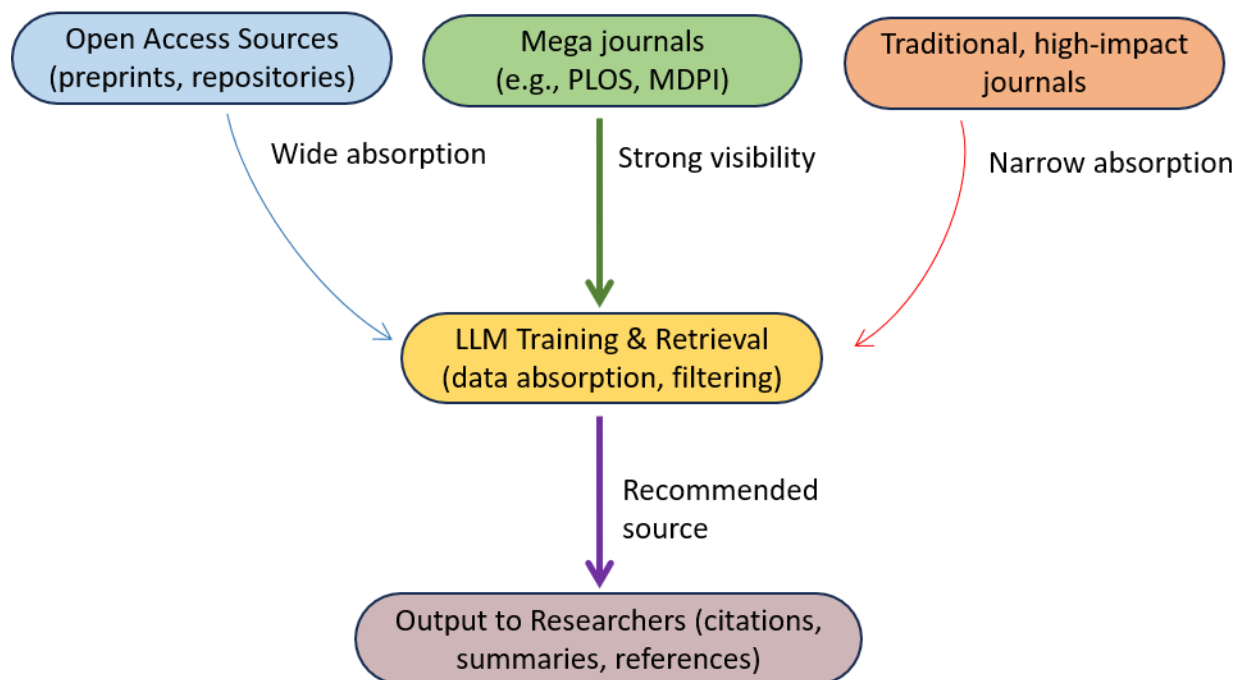


Figure 1: Citation visibility funnel: showing how publication visibility narrows through stages of source accessibility, absorption, and author credibility, with bias potentially amplifying at each layer

The funnel highlights how citation practices are shaped not only by content quality but also by systemic filters. Limited visibility of outlets, selective absorption by indexing platforms, and credibility biases toward certain author demographics can collectively skew citation patterns.

2.4 Prestige Effects and Democratization

When certain journals or authors repeatedly appear in GenAI-assisted discovery, users may conflate frequency with authority, mistaking algorithmic familiarity for academic endorsement. This could produce a form of prestige inflation—where inclusion in GenAI outputs signals legitimacy, regardless of peer-review rigor or scholarly lineage.

This shift also intersects with ongoing debates about the democratization of knowledge. On the one hand, the AI-facilitated amplification of OA sources could help level global disparities in research access, allowing scholars in lower-income institutions to engage with and contribute to cutting-edge debates. On the other hand, it raises concerns about epistemic quality: if citation practices increasingly depend on what is algorithmically accessible rather than what is most rigorous, the citation landscape may become distorted in ways that are difficult to detect or reverse.

In the next section, we explore how generative AI systems are built and used in scholarly contexts, with particular attention to the mechanisms that determine which sources are visible and which are excluded from the AI’s outputs.

3 AI-Mediated Visibility Bias in Scholarly Publishing

LLMs such as ChatGPT, Claude, and Gemini are trained on vast corpora of text data, drawn primarily from the open web. These corpora typically include publicly available websites, open-access repositories, Wikipedia, open code repositories, and digitized books or government documents. Due to copyright and licensing restrictions, subscription-based journal content—such as that found in Elsevier’s ScienceDirect, SpringerLink, or JSTOR—is largely excluded from training datasets unless explicitly licensed. Consequently, the internal knowledge embedded in these models reflects what is visible, open, and indexed online, rather than what is necessarily most authoritative or prestigious.

This visibility bias introduces important structural constraints on how generative AI constructs and conveys knowledge. When an AI system is asked to summarize a topic or suggest references, it draws on what it has seen. If an article was not in the training data, it does not exist from the model’s perspective. Even in models with retrieval-augmented generation (RAG)—where real-time web or database search complements internal memory—access is still limited by indexing, paywalls, and data-sharing agreements. Tools like Bing Chat or Perplexity.ai, which use web-based retrieval, will often cite sources that are open-access by default, since these are what the systems can legally and technically access in full.

In academic workflows, this bias is compounded by user behavior. Researchers using LLMs to assist with literature reviews or paper drafting may not question the provenance of suggested references, especially if the citations appear plausible and match their topic. In some cases, users may copy references directly from AI outputs, trusting that the model has already filtered the most relevant or authoritative sources. However, because visibility rather than rigor governs what the model can cite, there is a risk that citation practices become skewed toward those journals and platforms that are most accessible—not necessarily the most methodologically sound or peer-reviewed.

This mechanism is distinct from but analogous to the way Google Scholar and other search engines shape scholarly discovery. Prior research has shown that users tend to cite what appears near the top of search results, often influenced by factors like PDF availability, keyword density, or site optimization (Beel & Gipp, 2009; Jamali & Asadi, 2010). With generative AI, the issue is more opaque: the internal processes determining what content is surfaced are not only non-transparent but largely untraceable. The user cannot easily assess which training documents informed a particular claim or reference unless the system provides explicit citations—and even then, hallucinations or mismatches are common. Algorithms are not neutral selectors of information; they embody social, economic, and technical assumptions that shape what becomes visible or authoritative in digital systems (Kitchin, 2017).

Visibility bias in AI thus creates a new kind of epistemic filter. Sources that are well-indexed, freely available, and heavily linked are more likely to be incorporated into model outputs.

Sources behind paywalls, written in non-dominant languages, or published in low-visibility venues are less likely to be encountered—even if they contain valuable insights. The net result is a reshaping of the scholarly knowledge landscape, in which certain voices and venues are amplified algorithmically while others are suppressed or erased by omission. Over time, this may not merely bias citation counts, but recalibrate what counts as credible knowledge—reshaping academic canons based on algorithmic accessibility rather than intellectual merit.

This bias may also reinforce linguistic and regional hierarchies. Since most training data is English-language and heavily sourced from U.S. and European OA platforms, non-Western scholarship—especially from under-indexed or regional journals—may be further marginalized in AI-mediated outputs, even when methodologically sound.

Moreover, this phenomenon intersects with the broader dynamics of platform capitalism (Srnicek, 2016). Academic publishers that prioritize discoverability—by emphasizing high publication volumes, rapid turnaround times, search engine optimization, and open-access visibility—are particularly well positioned to benefit from this shift. Many have optimized their editorial and metadata practices to align with digital search systems, employing keyword-centric titles, expedited workflows, and indexing strategies designed to maximize exposure. These tactics resonate with the selection logics of generative AI systems, increasing the likelihood of citation through accessibility and algorithmic surfacing rather than through traditional markers of scholarly vetting or prestige. The epistemic authority of a journal article may thus become increasingly contingent not on peer review or citation lineage, but on algorithmic discoverability. Platform infrastructures increasingly mediate public knowledge flows and institutional legitimacy (Van Dijck et al., 2018)—positioning discoverability as a function of algorithmic optimization rather than peer review.

In this context, generative AI acts as both a research assistant and an epistemic gatekeeper. It mediates the discovery process and implicitly endorses certain sources over others, based on structural biases in its training and retrieval systems. As such, it plays a constitutive role in shaping scholarly knowledge—not only in what is cited, but in what is even seen. These dynamics echo broader critiques of big data epistemologies, which caution that the appearance of comprehensiveness often masks deep structural exclusions (boyd & Crawford, 2012).

The next section examines how this shift interacts with the rise of OA megajournals and the growing influence of predatory publishing, particularly as they intersect with AI-driven citation practices.

4 Research Directions in the Age of Generative Citation

4.1 Open Access Megajournals and Algorithmic Visibility

The structural visibility bias in generative AI intersects with broader trends in academic publishing, particularly the rapid expansion of OA megajournals (Heneberg, 2019; Spezi et al., 2017) and the ongoing concern about predatory publishing (Eriksson & Helgesson, 2017; Mills & Inouye, 2021). These developments complicate the relationship between openness, quality, and credibility—especially in a citation ecosystem increasingly shaped by algorithmic intermediaries.

OA megajournals—such as *Scientific Reports*, *PLOS ONE*, *Sustainability*, and *IEEE Access*—publish large volumes of articles across a wide range of disciplines, often with relatively short review times and broad criteria for acceptance (Björk, 2021; Wakeling et al., 2019). Their content is freely accessible, well-indexed, and optimized for online discovery. As such, these journals are heavily represented in search engines and digital repositories, and their articles are frequently encountered by users—and by AI systems trained or retrieving from the open web.

While many megajournals maintain reasonable peer-review standards, their scale and speed have prompted debates about editorial rigor and gatekeeping. More importantly, in the current context, their ubiquity and openness make them especially visible to LLMs. When generative AI tools draw on web-based corpora or use API-based search tools that prioritize OA content, megajournals become disproportionately influential. In the absence of prestige or impact-factor filtering, an AI system may recommend an article from an OA journal simply because it matches the query and is openly available—even if an equally relevant, higher-quality article exists behind a paywall.

4.2 Predatory Publishing and Citation Contamination

Yet the same conditions that elevate reputable megajournals also leave the door open to more problematic actors (Eriksson & Helgesson, 2017; Mills & Inouye, 2021). Predatory journals operate with minimal or nonexistent peer review, often charge high article processing fees, and exploit the OA model for profit (Beall, 2017; Kendall, 2021). While the definition of what a predatory journal is varies (Cobey et al., 2018), they are typically not indexed in reputable databases like Scopus or Web of Science, but some have succeeded in being listed in aggregators or gaining visibility through academic search engines (Shen & Björk, 2015). If such content is accessible on the open web, there is a nontrivial risk that it may be scraped into AI training data or retrieved during search-based generation.

Generative AI systems, unless specifically trained to distinguish between reputable and predatory sources, have no innate mechanism for quality control. If an LLM suggests a reference to a plausible-sounding article from a questionable journal, and a user includes it uncritically in their work, this introduces a form of citation contamination. Worse, as such references accumulate,

they may lend undeserved legitimacy to dubious outlets, reinforcing their presence in the scholarly record and potentially distorting future AI outputs that weight frequency or co-citation.

This risk is not hypothetical. Some publishers and professional organizations have already expressed concern about the amplification of poor-quality content via AI. For example, Finland's national research evaluation framework (JUFO) downgraded over 200 OA journals, citing concerns about volume over rigor (Kincaid, 2024). Studies such as Kendall and Teixeira da Silva (2024) warn that generative AI may exacerbate the influence of low-quality or fraudulent outlets by inadvertently recommending their content. Various reports of AI tools hallucinating citations to non-existent articles or fabricating journal titles further underscore the fragility of this ecosystem (Chelli et al., 2024; Puyt & Madsen, 2024). Further work is needed to distinguish between citation bias and citation fabrication—understanding how frequently users integrate hallucinated or non-existent references into formal academic work, and how such errors propagate within digital citation networks (Rekdal, 2014).

From a scientometric perspective, this introduces a new axis of distortion. Traditionally, citation analysis has grappled with issues like self-citation, citation cartels, and field-normalized variance. With AI in the loop, a new variable emerges: algorithmic citation bias—where the visibility and accessibility of content, rather than its scholarly merit, drives its inclusion in future works.

To mitigate this, developers of academic AI tools must incorporate quality filters, such as excluding known predatory journals, prioritizing DOAJ-listed publications, or flagging potentially unreliable sources (cf. Moher et al., 2017; Teixeira da Silva & Tsigaris, 2018). However, such safeguards are still emerging, and their implementation varies widely across platforms. In the interim, researchers must assume greater responsibility for verifying AI-generated references and maintaining rigorous citation standards.

Ultimately, the combination of algorithmic visibility, open-access scalability, and the porous boundary between credible and questionable OA publishing presents a complex challenge. Generative AI has the potential to both democratize and degrade scholarly citation practices. Whether it acts as a leveling force or a conduit for epistemic erosion will depend on the extent to which human users, institutional gatekeepers, and AI developers respond with informed vigilance.

The next section examines how these dynamics can be empirically studied and conceptually framed, and presents propositions for future research on the relationship between generative AI and scholarly citation behavior.

5 Future Research Agenda

5.1 Conceptual Propositions: Clusters and Mechanisms

Understanding the long-term impact of generative AI on academic citation practices requires a combination of conceptual clarity and empirical investigation. While this paper has outlined plausible mechanisms through which AI may amplify OA visibility and inadvertently legitimize questionable publications, the field remains nascent. In this section, we identify key areas for future research and propose conceptual propositions that can guide empirical studies.

A first priority is to empirically track whether generative AI use correlates with measurable shifts in citation behavior—particularly favoring OA sources and non-traditional venues. Studies could examine whether OA journals experienced a relative increase in citation rates after 2023, controlling for field, journal age, and publication volume. Comparing fields with (likely) high AI uptake among researchers (e.g., computer science, biomedicine) to those with lower uptake (e.g., history, philosophy) could provide further nuance.

Another direction involves experimental studies of citation behavior. Researchers could conduct controlled studies in which participants are tasked with completing literature reviews or writing summaries, either independently or with the assistance of AI tools. Analysis of the resulting reference lists would offer insights into how AI guidance influences the choice of sources, the diversity of cited journals, and the inclusion of low-prestige or open-access venues.

A third area concerns the development and testing of algorithmic safeguards. Developers of academic AI systems may implement filters or reputational scoring mechanisms to suppress low-quality sources and highlight peer-reviewed content. Researchers could test the efficacy of such filters in preventing the inclusion of predatory or unreliable references and assess whether users are more likely to trust and cite AI-recommended sources when transparency features (such as confidence scores or journal rankings) are included.

To guide future empirical and theoretical work on generative AI's impact on citation behavior, we propose seven interrelated propositions. These are grouped into three thematic clusters reflecting core dynamics: algorithmic visibility and discovery, user behavior and citation decision-making, and system-level risks and inequalities. Each proposition highlights a mechanism by which generative AI may alter the norms, patterns, and epistemic foundations of scholarly communication.

These propositions are not only empirically testable but also conceptually grounded in the literatures on information access, algorithmic mediation, and the sociology of knowledge production. They reflect a hybrid of visibility bias (Benkler, 2006), algorithmic filtering (Gillespie, 2014; Pariser, 2011), and cumulative advantage mechanisms such as the Matthew Effect (Merton, 1968)—updated for the age of machine-generated knowledge.

The first cluster concerns algorithmic visibility and discovery bias. These propositions address how large language models (LLMs) structurally favor open-access (OA) content in both training and retrieval, thereby reshaping which sources are seen, surfaced, and suggested in scholarly workflows.

Proposition 1: *The epistemic credibility of cited sources will increasingly depend on algorithmic visibility rather than peer-review provenance, unless deliberate interventions are made in AI training and output curation.*

As users increasingly rely on AI to find and generate references, sources that are more visible to the algorithm—regardless of methodological rigor—may come to dominate citation practices. This trajectory raises concerns about the long-term construction of scholarly legitimacy.

Proposition 2: *Generative AI systems amplify the OA citation advantage by disproportionately recommending sources that are freely accessible and indexed in open web corpora.*

Because LLMs are trained on publicly available content, they are more likely to “know about” and suggest OA articles. This creates an automated form of the OA citation advantage, independent of user intent or scholarly quality.

Proposition 3: *The extent of algorithmic citation bias in GenAI outputs is contingent on design decisions made by AI developers—particularly training corpus selection, source attribution transparency, and citation retrieval infrastructure.*

Visibility is not neutral; it is engineered. Licensing agreements, dataset composition, and interface affordances all shape what sources GenAI tools prioritize or exclude.

The second cluster focuses on researcher behavior and citation decision-making. These propositions emphasize how scholars interact with AI tools in practice, and how this affects which journals and articles are ultimately cited.

Proposition 4: *Researchers who use generative AI tools for literature discovery are more likely to cite lower-ranked or non-traditional journals than those who rely on conventional search methods.*

AI-assisted discovery lowers the cognitive and technical barriers to citing lesser-known outlets. While this may increase citation diversity, it also challenges traditional hierarchies of prestige and raises questions about quality assurance.

Proposition 5: *The presence of generative AI in academic workflows increases the citation share of high-volume OA megajournals, regardless of their impact factor or review standards.*

Megajournals such as *PLOS ONE*, *Scientific Reports*, and *Sustainability* publish at scale, are well-indexed, and highly visible in AI-generated outputs. As a result, they are more likely to be recommended and cited, potentially gaining disproportionate influence in AI-mediated citation ecosystems.

The third cluster addresses systemic risks and epistemic inequalities introduced by AI-mediated citation practices. These propositions highlight how the interaction of algorithmic visibility and openness can inadvertently amplify low-quality publications and widen disparities between fields and regions.

Proposition 6: *In the absence of filtering mechanisms, generative AI tools increase the risk of citation contamination by including references to predatory or low-rigor journals.*

Without curated citation engines or embedded reliability scoring, GenAI tools may recommend content from questionable journals simply because the content is openly accessible and matches the prompt.

Proposition 7: *Academic fields with a higher proportion of OA content will experience greater shifts in citation patterns under AI mediation than fields dominated by closed-access sources.*

AI-driven citation amplification is therefore likely to be uneven across disciplines. Fields like computer science, biomedicine, and education—where OA is more prevalent—may see greater impact than fields like philosophy or history, where key sources often remain behind paywalls.

Taken together, these seven propositions offer a multi-level framework for studying how generative AI affects scholarly citation practices. They encompass structural effects (bias in training data and retrieval systems), behavioral effects (how researchers engage with and trust AI outputs), epistemic consequences (prestige dilution and contamination risk), and systemic inequalities (disciplinary and global disparities in visibility). This conceptual agenda invites both empirical investigation and normative reflection on the future of scholarly communication in an algorithmically mediated research environment.

5.2 Research Roadmap

To synthesize the seven propositions into a structured agenda, Figure 2 presents a roadmap that organizes future research into three clusters: algorithmic visibility, researcher behavior, and system risks & inequalities. The roadmap displays three clusters of future directions—algorithmic visibility, researcher behavior, and system risks & inequalities—each branching into key propositions for advancing citation ethics in the age of generative AI.

Research Roadmap

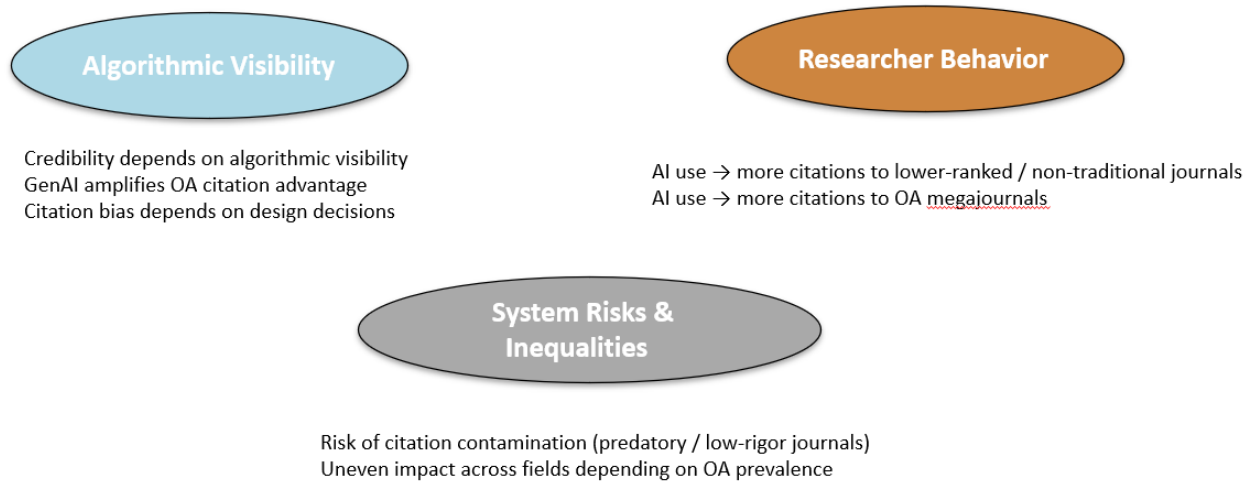


Figure 2: Research roadmap: three clusters of future directions—algorithmic visibility, researcher behavior, and system risks & inequalities—each branching into key propositions for advancing

The roadmap clusters the propositions into actionable domains. Algorithmic visibility addresses data and model transparency, researcher behavior emphasizes decision-making processes in citation, and system risks & inequalities capture predatory publishing and regional/language imbalances. Together, they provide a clear research agenda for empirical testing and conceptual refinement.

The concluding section reflects on the broader implications of this shift and considers how stakeholders might navigate a citation ecosystem increasingly mediated by algorithmic systems.

6 Conclusion: Canon, Credibility, and the Citation Future

Generative AI represents a powerful new intermediary in the academic knowledge ecosystem—one that influences not only how researchers write and think, but also what they see and cite. This paper argues that integrating large language models into scholarly workflows has the potential to significantly amplify the OA citation advantage, while also introducing new risks to the integrity of citation practices. The algorithmic architectures underlying generative AI tools are not neutral; they are shaped by data availability, indexing choices, and licensing constraints. As a result, OA content—regardless of quality—is far more likely to be encountered, summarized, and cited through AI-mediated discovery.

This shift carries both promise and peril. On one hand, it could help democratize access to research, increase visibility for under-cited authors and institutions, and expand the reach of

knowledge across linguistic and disciplinary boundaries. On the other hand, it may entrench new forms of epistemic distortion, where visibility substitutes for credibility and convenience undermines rigor. Particularly troubling is the potential for low-quality or predatory publications to gain traction in citation networks via AI amplification, especially in fields with high publication volume and limited gatekeeping.

The algorithmic mediation of citation risks reinforcing long-standing North-South divides in scholarly communication (Collyer, 2018; Lor, 2007). Research produced in low-resource or non-English contexts—already under-indexed in major databases—may face a second-order erasure by GenAI systems trained on predominantly Western and Anglophone corpora. Unless deliberate efforts are made to include diverse linguistic and regional content, citation visibility will remain structurally skewed.

The academic community faces a critical inflection point. Researchers, developers, publishers, and institutions must acknowledge that citation practices are no longer solely a matter of human discernment—they are now mediated by opaque systems trained on imperfect and often biased corpora. The implications of this transition are far-reaching. They touch not only on scholarly impact and evaluation, but on the construction of knowledge itself: what is seen, remembered, and passed on.

As AI shapes what is seen and cited, it may also shape what is remembered. Over time, algorithmic citation patterns may contribute to the canonization of specific journals or ideas—not based on intellectual lineage, but on data availability and digital visibility. This process risks substituting legacy with latency: what is frequently surfaced becomes foundational, even if it lacks sustained scholarly engagement. In the long run, algorithmic citation trails may shape collective memory in ways that outpace traditional mechanisms of academic consensus or peer-reviewed recognition.

To navigate this landscape responsibly, several strategies are needed. First, researchers must develop a critical understanding of how AI tools generate suggestions and be trained to verify and contextualize any references provided. Second, academic institutions and publishers should support the development of transparent and accountable AI tools that prioritize reputable sources and disclose uncertainty or provenance. Third, bibliometric research must closely monitor shifts in citation behavior over time, paying particular attention to changes in source diversity, prestige distribution, and the emergence of new citation norms.

If approached reflexively and collaboratively, the integration of generative AI into scholarly communication can be harnessed to enhance the visibility of rigorous and inclusive research. If left unchecked, however, it may erode long-standing mechanisms for ensuring quality, accountability, and epistemic trust. Citations are more than academic footnotes—they are a record of intellectual lineage and legitimacy. In the age of algorithmic authorship, safeguarding the integrity of that record is no longer optional—it is a shared responsibility across scholars, developers, and institutions.

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